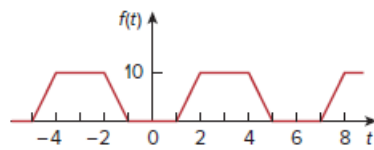


Problem

Find the Fourier series for the signal in Fig. 17.58. Evaluate $f(t)$ at $t = 2$ using the first three nonzero harmonics.

Figure 17.58:



Step-by-step solution

Step 1 of 7

Refer to waveform in Figure 17.58 in the textbook.

The function is described as,

$$f(t) = \begin{cases} 0 & \text{for } 0 < t < 1 \\ 10(t-1) & \text{for } 1 < t < 2 \\ 10 & \text{for } 2 < t < 4 \\ -10(t-5) & \text{for } 4 < t < 5 \\ 0 & \text{for } 5 < t < 6 \end{cases}$$

Calculate the fundamental frequency.

$$\omega_0 = \frac{2\pi}{T}$$

Substitute 6 for T in the above equation.

$$\begin{aligned} \omega_0 &= \frac{2\pi}{6} \\ &= \frac{\pi}{3} \end{aligned}$$

Step 2 of 7

Calculate the value of coefficient a_0 .

$$\begin{aligned}
 a_0 &= \frac{1}{T} \int_0^T f(t) dt \\
 &= \frac{1}{6} \int_0^6 f(t) dt \\
 &= \frac{1}{6} \left[\int_0^1 (0) dt + \int_1^2 10(t-1) dt + \int_2^4 10 dt + \int_4^5 10(5-t) dt + \int_5^6 (0) dt \right] \\
 &= \frac{1}{6} \left[0 + (10) \left(\frac{t^2}{2} - t \right)_1^2 + (10t)_2^4 + (10) \left(5t - \frac{t^2}{2} \right)_4^5 + 0 \right] \\
 &= \frac{1}{6} \left[0 + (10) \left(\frac{2^2}{2} - 2 - \frac{1^2}{2} + 1 \right) + 10(4-2) + (10) \left(5(5) - \frac{5^2}{2} - 5(4) + \frac{4^2}{2} \right) \right] \\
 &= \frac{1}{6} [(10)(0.5) + 20 + (10)(0.5)] \\
 &= \frac{1}{6} (30) \\
 &= 5
 \end{aligned}$$

Step 3 of 7

Calculate the value of coefficient a_n .

$$\begin{aligned}
 a_n &= \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t) dt \\
 &= \frac{2}{6} \int_0^6 f(t) \cos(n\omega_0 t) dt \\
 &= \left[\frac{1}{3} \int_0^1 f(t) \cos(n\omega_0 t) dt + \frac{1}{3} \int_1^2 f(t) \cos(n\omega_0 t) dt + \frac{1}{3} \int_2^4 f(t) \cos(n\omega_0 t) dt \right. \\
 &\quad \left. + \frac{1}{3} \int_4^5 f(t) \cos(n\omega_0 t) dt + \frac{1}{3} \int_5^6 f(t) \cos(n\omega_0 t) dt \right] \\
 &= \left[\frac{1}{3} \int_0^1 0 \cdot \cos(n\omega_0 t) dt + \frac{1}{3} \int_1^2 10(t-1) \cdot \cos(n\omega_0 t) dt + \frac{1}{3} \int_2^4 10 \cdot \cos(n\omega_0 t) dt \right. \\
 &\quad \left. + \frac{1}{3} \int_4^5 -10(t-5) \cos(n\omega_0 t) dt + \frac{1}{3} \int_5^6 0 \cdot \cos(n\omega_0 t) dt \right] \\
 &= \frac{1}{3} \int_1^2 10(t-1) \cdot \cos(n\omega_0 t) dt + \frac{1}{3} \int_2^4 10 \cdot \cos(n\omega_0 t) dt + \frac{1}{3} \int_4^5 -10(t-5) \cos(n\omega_0 t) dt \\
 &= \frac{1}{3} \int_1^2 10(t-1) \cdot \cos\left(\frac{n\pi}{3} t\right) dt + \frac{1}{3} \int_2^4 10 \cdot \cos\left(\frac{n\pi}{3} t\right) dt + \frac{1}{3} \int_4^5 -10(t-5) \cos\left(\frac{n\pi}{3} t\right) dt
 \end{aligned}$$

Step 4 of 7

On further simplification:

$$\begin{aligned}
 a_n &= \left[\frac{10}{3} \cdot \frac{3}{n\pi} \left[(t-1) \sin \left(\frac{n\pi}{3} t \right) \right]_{t=1}^{t=2} + \frac{30}{n^2 \pi^2} \left[\cos \left(\frac{n\pi}{3} t \right) \right]_{t=1}^{t=2} + \frac{10}{3} \cdot \frac{3}{n\pi} \left[\sin \left(\frac{n\pi}{3} t \right) \right]_{t=2}^{t=4} \right. \\
 &\quad \left. - \frac{10}{3} \cdot \frac{3}{n\pi} \left[(t-5) \sin \left(\frac{n\pi}{3} t \right) \right]_{t=4}^{t=5} - \frac{30}{n^2 \pi^2} \left[\cos \left(\frac{n\pi}{3} t \right) \right]_{t=4}^{t=5} \right] \\
 &= \left[\frac{10}{n\pi} \sin \left(\frac{2n\pi}{3} \right) + \frac{30}{n^2 \pi^2} \left[\cos \left(\frac{2n\pi}{3} \right) - \cos \left(\frac{n\pi}{3} \right) \right] + \frac{10}{n\pi} \left[\sin \left(\frac{4n\pi}{3} \right) - \sin \left(\frac{2n\pi}{3} \right) \right] \right. \\
 &\quad \left. - \frac{10}{n\pi} \sin \left(\frac{4n\pi}{3} \right) - \frac{30}{n^2 \pi^2} \left[\cos \left(\frac{5n\pi}{3} \right) - \cos \left(\frac{4n\pi}{3} \right) \right] \right] \\
 &= \frac{30}{n^2 \pi^2} \left[\cos \left(\frac{2n\pi}{3} \right) - \cos \left(\frac{n\pi}{3} \right) \right] - \frac{30}{n^2 \pi^2} \left[\cos \left(\frac{5n\pi}{3} \right) - \cos \left(\frac{4n\pi}{3} \right) \right] \\
 &= \frac{30}{n^2 \pi^2} \left[\cos \left(\frac{2n\pi}{3} \right) - \cos \left(\frac{n\pi}{3} \right) \right] - \frac{30}{n^2 \pi^2} \left[\cos \left(\frac{n\pi}{3} \right) - \cos \left(\frac{2n\pi}{3} \right) \right] \\
 &= \frac{60}{n^2 \pi^2} \left[\cos \left(\frac{2n\pi}{3} \right) - \cos \left(\frac{n\pi}{3} \right) \right]
 \end{aligned}$$

Step 5 of 7

Since the function is even, $b_n = 0$.

Calculate the Fourier series.

$$\begin{aligned}
 f(t) &= a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \\
 &= 5 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + 0) \\
 &= 5 + \sum_{n=1}^{\infty} \left\{ \frac{60}{n^2 \pi^2} \left[\cos \left(\frac{2n\pi}{3} \right) - \cos \left(\frac{n\pi}{3} \right) \right] \cos \left(\frac{n\pi}{3} t \right) \right\}
 \end{aligned}$$

Therefore, the Fourier series, $f(t)$ for the signal is,

$$\boxed{5 + \sum_{n=1}^{\infty} \left\{ \frac{60}{n^2 \pi^2} \left[\cos \left(\frac{2n\pi}{3} \right) - \cos \left(\frac{n\pi}{3} \right) \right] \cos \left(\frac{n\pi}{3} t \right) \right\}}$$

Step 6 of 7

Using the first 3 nonzero harmonics, the value of function $f(t)$ at $t = 2s$ is approximated as,

$$a_1 = \frac{60}{(1)^2 \pi^2} \left[\cos\left(\frac{2\pi}{3}\right) - \cos\left(\frac{\pi}{3}\right) \right]$$

$$= -6.07927$$

$$a_2 = \frac{60}{(2)^2 \pi^2} \left[\cos\left(\frac{4\pi}{3}\right) - \cos\left(\frac{2\pi}{3}\right) \right]$$

$$= 0$$

$$a_3 = \frac{60}{(3)^2 \pi^2} [\cos(2\pi) - \cos(\pi)]$$

$$= 1.35095$$

$$a_4 = \frac{60}{(4)^2 \pi^2} \left[\cos\left(\frac{8\pi}{3}\right) - \cos\left(\frac{4\pi}{3}\right) \right]$$

$$= \frac{60}{(4)^2 \pi^2} \left[\cos\left(\frac{2\pi}{3}\right) - \cos\left(\frac{4\pi}{3}\right) \right]$$

$$= 0$$

$$a_5 = \frac{60}{(5)^2 \pi^2} \left[\cos\left(\frac{10\pi}{3}\right) - \cos\left(\frac{5\pi}{3}\right) \right]$$

$$= \frac{60}{(5)^2 \pi^2} \left[\cos\left(\frac{4\pi}{3}\right) - \cos\left(\frac{\pi}{3}\right) \right]$$

$$= -0.24317$$

Step 7 of 7

Now, calculate the Fourier series at $t = 2$.

$$\begin{aligned} f(2) &\approx 5 + \sum_{n=1}^5 \left\{ \frac{60}{n^2 \pi^2} \left[\cos\left(\frac{2n\pi}{3}\right) - \cos\left(\frac{n\pi}{3}\right) \right] \cos\left(\frac{2n\pi}{3}\right) \right\} \\ &= 5 + (-6.07927) \cos\left(\frac{2\pi}{3}\right) + (1.35095) \cos(2\pi) + (-0.24317) \cos\left(\frac{10\pi}{3}\right) \\ &= 5 + (-6.07927) \left(-\frac{1}{2}\right) + (1.35095) + (-0.24317) \left(-\frac{1}{2}\right) \\ &= 9.51217 \end{aligned}$$

Therefore, the approximate value of $f(t)$ at $t = 2$ is 9.51217.